

AI Evolution as an Advancing Approximation Frontier

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Abstract

AI development is often described as progress toward increasingly capable systems. This paper asks a different question: **how does successful progress change the development problems that remain?** It treats AI development as an advancing approximation process in which artificial systems increasingly reproduce, exceed, or substitute for functions associated with intelligent human performance through mechanisms that can differ substantially from those that generate human intelligence.

We develop a conditional account of frontier change. Differential resolution is one mechanism: when some development problems resolve faster than others, progress can change the **composition** of the unresolved frontier rather than simply reduce its size. This motivates the empirical hypothesis that dimensions with weakly specified or difficult-to-evaluate targets may resolve more slowly than dimensions with clearer targets and feedback. Other mechanisms can change the frontier through different routes: capability can outrun evaluation, human practices and evaluative standards can alter the reference conditions for later systems, and AI can increasingly participate in the development of successor systems.

These mechanisms also create a diagnostic problem. Developers may face several plausible explanations of a current limitation, each implying different interventions, while the system being diagnosed continues to change. We use Prediction–Attribution Conjugacy, developed in earlier work, to examine when such explanations are structurally distinguishable, when they are identifiable in practice, and when diagnostic learning may race against frontier movement.

The central contribution of the paper is an account of how AI progress can change the composition, visibility, reference conditions, and developmental structure of the problems that come next. Questions about a fixed AI ceiling or a particular architectural succession remain open.

Keywords: artificial intelligence; AI development; approximation; development frontier; evaluation lag; human–AI coevolution; prediction–attribution conjugacy

1. Introduction — Progress Changes the Problem That Remains

AI development has repeatedly changed what counts as the next difficult problem. Scaling-law work suggested that larger models, more data, and more compute could produce predictable gains. Compute-optimal training then showed that how a fixed compute budget was allocated could matter more than simply increasing parameter count. More recently, inference-time scaling has shown that how computation is used can matter as much as how much pretrained capacity a system has [1], [2], [3].

The development problem is therefore not simply the same problem becoming smaller.

This paper examines that observation as a general feature of AI development. We use **development frontier** to mean the set and structure of unresolved problems that materially constrain or shape further AI advancement. A current **bottleneck** is one binding constraint within that frontier. The frontier is broader than the research agenda: what researchers choose to work on is a response to the frontier, not the frontier itself.

We ask:

How does continued AI progress change the composition and character of the development problems that remain?

The question arises because progress can do more than remove deficiencies. Artificial systems may improve rapidly on some dimensions while other dimensions improve slowly. New capabilities can expose behaviors that earlier evaluations were not designed to see. Widespread AI use can change human practices, expectations, and evaluative standards. AI can also become part of the process that builds the next generation of AI. Each of these changes affects what the next development problem is.

We begin with approximation, then follow the consequences of uneven progress through the residual, evaluation, the human reference, and successor development. Once those elements can all move, the remaining analytical problem is diagnostic: how can developers distinguish among competing explanations of a limitation before the relevant structure changes again?

This paper contributes a conditional account of how successful AI development can change the composition and character of that frontier. We develop four mechanisms—differential resolution, evaluation lag, human-reference feedback, and AI-mediated successor development—and derive empirically examinable implications for which problems persist, when failures become visible, how development conditions change, and when frontier attributions remain identifiable.

2. Approximation Without Generative Equivalence

We begin with approximation because the central argument depends on the possibility that different dimensions of intelligent performance can improve through different mechanisms and at different rates.

Here, **approximation** means reproducing, exceeding, or substituting for functions associated with intelligent human performance. It concerns functional performance rather than reconstruction of human cognition. A compact notation for this relationship appears in Appendix A, but the main point does not depend on treating either human or artificial intelligence as a single scalar quantity.

The Transformer is a useful example. Vaswani et al. replaced the recurrent and convolutional structures then dominant in sequence modeling with an architecture based on attention alone. The architecture improved translation performance while permitting greater parallelization and faster training [5]. The advance came from a better artificial mechanism for performing language-related tasks rather than from a more complete theory of how humans generate language.

The example therefore establishes that improved task performance can arise from an artificial mechanism that does not reconstruct the human process that generated the corresponding capability. **Task approximation and generative equivalence are distinct.**

That distinction matters because improvement in an artificial system depends on the guidance available to its own development process. When the target is clear and feedback is informative, developers can compare candidate changes, identify deficiencies, and direct subsequent improvement without first reconstructing the corresponding human generative process. When adequacy depends on context, social role, institutional purpose, competing values, or changing expectations, the same improvement loop may provide weaker guidance because it is harder to determine what counts as a better result.

Different dimensions can therefore present different conditions for improvement even within the same overall system. This leads to the next question: if some dimensions are resolved more readily than others, how does that change the set of problems that remains?

3. Differential Resolution Changes the Residual

Technological progress often moves the bottleneck rather than eliminating the need for one. Histories of large technical systems describe “reverse salients”: lagging components whose resolution exposes the next critical problem in an advancing system [6]. AI development provides a contemporary version of that broader pattern.

Scaling illustrates the sequence. Early scaling-law work showed regular relationships among model size, data, compute, and loss [1]. Compute-optimal training then changed the practical diagnosis: under a fixed training budget, inefficient allocation between parameters and training tokens could be more limiting than raw scale. Chinchilla operationalized that revised diagnosis and outperformed substantially larger models under the same training-compute budget [2].

Sparse mixture-of-experts methods relaxed another constraint by activating only selected experts for each input, allowing effective capacity to rise without proportional growth in computation [7]. Inference-time scaling changed the location of productive computation again: Snell et al. showed that on some problems a smaller model using well-allocated test-time compute could outperform a much larger pretrained model under matched-compute conditions [3].

Taken together, these episodes establish the point needed here: **different constraints can become relatively important at different stages.**

By **residual**, we mean the set of human-relevant discrepancies that remain materially unresolved for the comparison or task under consideration. Its composition can change in at least two ways. Existing discrepancies can leave at different rates, and new discrepancies can enter as capabilities, contexts, or expectations change.

A residual discrepancy becomes part of the development frontier when it becomes consequential for useful deployment or further advancement. A system may perform well enough on one dimension that the remaining discrepancy no longer constrains development, while another discrepancy becomes newly binding

because the system is deployed more widely, used for a more demanding task, or evaluated against a higher standard. The residual and the frontier therefore overlap, but they are not identical: the residual describes unresolved human-relevant discrepancies, while the frontier identifies the unresolved problems that materially shape what development must address next.

Differential resolution is one mechanism by which that composition can change rather than a necessary condition for every kind of compositional change.

The empirical question pursued here is narrower: among discrepancies already under consideration, do resolution rates differ systematically with properties that matter?

We propose the following provisional hypothesis:

Well-specified and easily evaluated dimensions tend, over relevant periods, to be more tractable than dimensions whose targets are contested, contextual, weakly specified, or difficult to evaluate.

The distinction between **mechanism-side** and **target-side** difficulty helps explain why.

A mechanism-side problem asks how the artificial system can learn, route, store, or apply information more effectively. A target-side problem asks what behavior should count as adequate and how that adequacy can be established.

Consider two tasks under different evaluative conditions. Translating a sentence into French under an agreed task definition and scoring standard gives developers a relatively clear basis for comparing outputs. Summarizing a contract “fairly” without prior agreement about whose interests matter, which omissions are consequential, or what legal role the summary serves gives much weaker guidance about what improvement means. The contrast is not between translation and summarization as task categories; it is between development settings with clearer and less settled adequacy criteria.

Alignment research provides more demanding examples. A recent review of 94 influential value-alignment papers found that most did not explicitly define “values” and often relied on preferences as a proxy for more socially and culturally situated concepts [8]. Related social-choice work emphasizes that human feedback can contain genuine disagreement, making some alignment problems partly problems of aggregation rather than discovery of one latent preference [9].

Target-side difficulty can therefore arise from at least four sources: **specification, plurality, context, and evaluation**. These difficulties can slow improvement when they make feedback less discriminating: candidate changes may produce observable differences without making it clear which result is better, why it is better, or whether the improvement will survive a change of context. Better architectures, richer information, improved evaluation, or a future approximation regime may reduce these difficulties. The hypothesis is not that they are permanent. It is that, when these conditions weaken the improvement loop, some dimensions may leave the residual more slowly than others over relevant periods.

This hypothesis is testable. Evidence against it would include broad capability improvements that reduce difficult target-side discrepancies as fast as or faster than clearer dimensions, or longitudinal comparisons showing no relationship between specifiability, evaluability, and time to reliable performance. A prospective

test also has to avoid defining difficulty by the fact that something remained unresolved; Appendix B sets out a compact design for doing so.

The differential-resolution mechanism is a form of survivor selection. When members of an unresolved population leave at different rates, the composition of the survivor population changes. Related work develops this mechanism for unresolved inquiry [10]. Here, the population is the set of unresolved AI-development discrepancies, and the question is whether attributes such as specifiability and evaluability predict differential resolution.

This survivor-selection mechanism is distinct from **entry** into the residual. New capabilities, contexts, or expectations can introduce discrepancies that were previously irrelevant or invisible. Either process can change residual composition; the hypothesis in this section concerns the selection effect generated by unequal exit rates.

Progress can therefore reduce the amount of unresolved discrepancy while changing its composition. That establishes one way the frontier can change. The next issue is whether the problems that remain or newly enter are visible soon enough to shape development.

4. Evaluation Can Lag Capability

A changing residual matters differently depending on whether the remaining problems are readily observable. This section therefore moves from **what remains** to **what can be seen**.

Capability can advance faster than the evaluation regime designed for an earlier system. OpenAI reported in 2026 that a model operating over extended tasks exhibited failure modes during limited internal use that existing pre-deployment evaluations had not captured. The important change was that longer persistence created behavior requiring evaluation at the trajectory level rather than one action at a time [11].

That example illustrates a specific mechanism:

Hard-to-evaluate residual failures can persist unnoticed while capability, deployment, or autonomy expands around them.

The sequence is straightforward. A system gains a new capability or longer operating horizon. Existing evaluations continue to measure shorter or simpler behavior. A deficiency that was previously irrelevant or invisible remains undetected. Wider deployment then increases the exposure of that unresolved deficiency.

Residual selection and evaluation lag therefore answer different questions. Residual selection asks **which kinds of problems remain**. Evaluation lag asks **which remaining or newly enabled problems are visible under the current evaluation regime**.

The long-horizon example shows capability outrunning an existing evaluation regime and revealing a class of behavior that earlier evaluations missed. Residual concentration in hard-to-evaluate dimensions remains a separate proposition that depends on the differential-resolution hypothesis in Section 3.

The mechanism is also conditional rather than one-directional. Capability improvements can strengthen monitoring, self-correction, and safety research. The relevant proposition is narrower: when hard-to-evaluate dimensions also resolve more slowly, evaluation lag can allow them to persist while average performance improves and deployment expands.

So far, the development frontier has moved because artificial-system problems resolve unevenly and evaluation can lag new capability. The next step is to examine a second source of movement: the human reference against which later systems are interpreted and evaluated.

5. The Human Reference Can Move

The **human reference** is the evolving set of human capabilities, practices, expectations, and evaluative standards against which AI performance is interpreted. It is broader than any abstract representation of human intelligence. The relevant question here is whether changes in those reference conditions feed back into later AI development.

A long tradition of distributed and extended cognition treats tools and environments as potentially integral to cognitive activity rather than as passive accessories [12]. AI extends that familiar issue because it can participate across many stages of intellectual work rather than supporting only one fixed operation.

5.1 Assisted performance and practiced capability

Research on cognitive offloading establishes an important distinction: assisted performance is not the same as independent capability.

A 2026 review in *Trends in Cognitive Sciences* concluded that offloading cognition to AI can impede skill acquisition and contribute to skill decay in some settings, while emphasizing that outcomes depend strongly on how AI is used [13]. In a controlled logic-puzzle study, greater reliance on AI improved assisted performance but was associated with weaker subsequent unassisted performance, whereas independent reasoning during the assisted phase was associated with greater gains [14].

These studies establish a task-specific result: what people can do **with** AI and what they continue to practice or perform **without** AI can move differently.

That distinction matters for later AI development because humans remain part of evaluation, oversight, demand formation, and the production of training or feedback data. Changes in human practice can therefore alter the environment in which later systems are judged and improved.

5.2 Four feedback channels

The feedback becomes easier to examine when it is separated into concrete channels.

- **Evaluation and oversight:** the capabilities of human evaluators affect what errors are detected and what behavior is rewarded.
- **Training and feedback data:** human practices altered by AI can change the data later systems learn from.

- **Demand and expectations:** widespread use changes what users and institutions ask later systems to do.
- **Division of labor:** automation changes which human contributions remain scarce and therefore which bottlenecks matter.

These channels are the basis for treating the human reference as partly endogenous to prior AI use. The relevant changes can occur in practice, distribution, expectations, and institutional roles without requiring rapid biological change.

Because those changes alter the human side of successor development, the argument now moves to the other side of the same process: how AI itself enters the production of the next generation of AI.

6. When AI Participates in AI Development

AI systems are increasingly used inside AI research and engineering. They write code, generate synthetic data, conduct evaluations, search literature, propose experiments, analyze results, and assist with research workflows.

Anthropic reported in 2026, using one internal measurement framework, that Claude “led” 26% of measured AI research and engineering work and collaborated substantially on more than 90%, while not operating fully autonomously in any measured subset [15]. The figure is a single-lab illustration rather than a cross-industry measure, but it establishes that substantial AI participation in frontier development is already possible.

The development relation is therefore increasingly mixed:

$$\text{Human} + \text{AI}_t \rightarrow \text{AI}_{t+1}$$

Later systems can inherit substantive developmental contributions produced by earlier systems while humans retain substantial control over objectives, evaluation criteria, and deployment.

6.1 Variation and selection are different functions

AI participation becomes analytically clearer once **variation** is separated from **selection**.

AI may increasingly generate candidate code, experiments, training procedures, architectures, or data without controlling what counts as improvement. Selection can then be decomposed into three functions:

1. **criterion setting** — deciding what counts as better;
2. **evaluation execution** — measuring candidates against those criteria;
3. **successor decision** — deciding which system or method actually advances.

AI can perform parts of evaluation execution in benchmarked settings while humans or institutions retain substantial authority over criterion setting and successor decisions. Markets, users, regulation, infrastructure, and capital allocation also impose selection pressures that are neither purely human-developer nor purely AI-generated.

The distinction matters because machine-generated variation can expand without a corresponding transfer of authority over what counts as success.

This yields a developmental-lineage proposition:

Later AI systems can become increasingly machine-derived in their developmental inputs while human or institutional authority remains substantial in criterion setting, evaluation, and successor selection.

6.2 Machine-derived contribution can improve or degrade development

The lineage proposition says where developmental inputs can come from; it does not by itself say whether those inputs improve the successor system. That depends on how machine-generated material is produced, tested, and incorporated.

AI-run experiments, adversarial cases, externally verified search results, and discovered optimization strategies can add useful information. Recursive reuse of model-generated outputs can do the opposite. Shumailov et al. showed that indiscriminate recursive training on generated data can produce model collapse and loss of information about the original distribution [16].

The relevant distinction is therefore between machine-generated material that is independently tested or informative and material that mainly recycles earlier approximation error.

6.3 Automation can move the human bottleneck

AI participation in R&D also returns us to differential resolution.

As coding, experiment execution, search, and benchmark-driven optimization become easier to automate, remaining human work may become more concentrated in tasks such as problem choice, interpretation of ambiguous evidence, objective setting, governance, and tradeoff judgment.

Whether this concentration occurs is an empirical question. Some judgment-intensive tasks may also become highly automatable. The relevant test is whether the human share of AI development becomes increasingly concentrated in weakly specified and judgment-intensive functions as more structured work is automated.

A moving frontier, a changing human reference, and increasing AI participation in successor development together create a new problem: developers must decide what to try next even when several explanations of the current limitation remain plausible. We use Prediction–Attribution Conjugacy (PAC) to examine that diagnostic problem.

7. Diagnosing a Moving Frontier

7.1 The diagnostic problem

Suppose a persistent deficiency could be attributed to architecture, data, compute allocation, inference, evaluation, or a changing target. These explanations matter because they imply different interventions. The

practical problem is not merely to list them. It is to determine whether available evidence can distinguish among them before the relevant development context changes.

7.2 The PAC framework and its source

PAC was developed in earlier work as a framework for adaptive decision problems in which competing attributions imply different prospective outcomes and therefore different actions [4]. Competing explanations may imply different outcomes under a suitable test. To guide a development decision, however, developers must also be able to conduct an informative test and obtain its results while the distinction remains relevant.

PAC names these two conditions separately.

Structural conjugacy exists when competing decision-relevant attributions imply different prospective outcomes under at least one common informative condition.

Identified conjugacy requires feasible observations or interventions to provide enough diagnostic variation to distinguish those accounts in time for the decision.

The framework comes from the earlier PAC paper [4]. The application here is to the AI development frontier, where the object being diagnosed can change while evidence is being gathered.

A compact representation is:

$$A_i \rightarrow P_i \rightarrow D_i$$

where an attribution A_i generates a prospective implication P_i , which motivates an intervention D_i .

7.3 Applying PAC to AI development

Chinchilla is a relatively favorable case. A compute-allocation attribution generated a concrete prospective test: under a fixed compute budget, a smaller model trained on substantially more data should outperform a larger undertrained one. The resulting experiment provided meaningful diagnostic evidence [2].

Other frontier questions are less identified. Large training runs often bundle architecture, data, compute, post-training, tools, and evaluation changes. Even when an intervention succeeds, several explanations may remain compatible with the outcome.

The same issue arises when developers ask whether the broad approximation mechanism itself has become binding. An **approximation-regime change**, as used here, means a materially different approximation mechanism whose gains cannot be reproduced by reasonable variations within the existing broad mechanism under comparable constraints. That is a judgment category rather than a sharply observable boundary.

If the current broad mechanism is binding, a materially different mechanism should eventually produce gains that continued within-regime adjustments cannot. But compute, data, inference, and system composition remain rival explanations. The diagnostic challenge is therefore to design evidence that discriminates among these accounts rather than merely recording another performance gain.

PAC helps formulate that challenge without forecasting which development path will prevail.

7.4 The identification race

The application becomes especially important when the frontier moves while developers are trying to understand it.

The PAC paper describes an exploratory **identification race** using the conceptual ratio

$$\rho = \lambda_i / \lambda_r$$

where λ_i is the rate at which evidence reduces uncertainty among competing attributions and λ_r is the rate at which the relevant attributional structure changes [4].

If diagnostic learning substantially outruns structural change, understanding can stabilize. If the two proceed on similar timescales—or structural change is faster—frontier diagnoses can remain provisional.

The rate of structural change varies across AI-development problems. Some relationships, such as loss–compute scaling within a stable training regime, may persist long enough for diagnostic learning to accumulate. Other parts of the frontier, including agentic behavior, deployment conditions, tool use, or evaluation regimes, can change more rapidly.

Our application is specific: PAC asks whether competing explanations of an AI-development frontier are identifiable **before the relevant structure changes enough to weaken the value of the old distinction**.

As AI research agents take on more of the development loop, attribution, prediction, intervention, and system change may also become increasingly compressed in time. That possibility connects the diagnostic problem back to the broader claim that progress can change the developmental process itself.

8. The Frontier Is Partly Endogenous

The preceding sections can now be assembled into one account.

The **development frontier** is the structure of unresolved problems that materially shape further AI advancement. Its next state can be influenced by four mechanisms examined here:

1. **Differential resolution:** some problems are resolved faster than others, changing residual composition.
2. **Evaluation lag:** new capability can expose problems that earlier evaluation regimes did not detect.
3. **Human-reference change:** prior AI use can change evaluation, data, expectations, and division of labor.
4. **AI-mediated development:** earlier AI can contribute directly to the production and evaluation of successor systems.

The mechanisms can also interact. AI-assisted work can change what humans practice and how they evaluate outputs; those changes can alter the feedback supplied to successor systems; successor systems can then encounter a different set of discrepancies under a different evaluation regime. Differential resolution changes **what remains**, evaluation lag changes **what is visible**, human-reference feedback can change **what counts**

as adequate, and AI-mediated development changes **how the next system is produced**. Their combination therefore affects the content, observability, evaluative reference, and production process of the frontier at the same time.

This is the sense in which the frontier is **partly endogenous to previous progress**. Progress changes the system, the environment in which the system is judged, and increasingly the process that produces the next system.

If this account captures an important part of frontier movement, we should expect several empirical patterns: lower specifiability or evaluability should sometimes predict slower resolution among otherwise comparable problems; newly enabled capabilities should sometimes reveal failure classes not well measured by prior evaluations; changes in human practice should feed into later evaluation, data, demand, or oversight; and increasing AI participation in R&D should sometimes change the composition of remaining human work.

The frontier is only partly endogenous because AI development also faces constraints supplied from outside these mechanisms: chip supply, energy, capital, data access, regulation, institutional choices, market demand, and geopolitical conditions. Any of these can dominate at a particular time.

The account's explanatory reach should narrow when the expected patterns repeatedly fail to appear—for example, when prospectively coded dimensions with lower specifiability or evaluability do not resolve more slowly than otherwise comparable clearer dimensions, when human-practice changes have little effect on later development conditions, when increasing AI participation leaves the composition of human work unchanged, or when evaluation systems reliably detect newly enabled failure classes before or at deployment. Its reach should also narrow when careful case analysis explains frontier movement better through externally supplied constraints.

These conditions follow directly from the mechanisms and define their explanatory reach. The account earns explanatory weight only to the extent that those mechanisms are present.

9. Conclusion

Our motivating concern is simple: **successful AI development can change the problem that developers face next**.

The approximation view explains why. Artificial systems can improve through mechanisms that differ from the processes they approximate. Differential resolution then allows some discrepancies to disappear faster than others, changing the composition of what remains. Evaluation lag can delay recognition of newly important failures. AI use can alter the human practices and standards that shape later evaluation. AI participation in R&D can change the process through which successor systems are produced. Once those elements move together, the diagnosis of the frontier can itself become time-sensitive.

The familiar question is:

How close can AI come to human intelligence?

We add a different set of questions:

What kinds of development problems remain after each major advance? Which resolve more slowly? Which become visible only after capability expands? How does the human reference change? And can competing explanations of the next bottleneck be identified before the frontier changes again?

Whether a fixed capability ceiling exists remains open. A new approximation regime may resolve problems that look difficult today, and external constraints may dominate particular stages of development. The account remains compatible with those possibilities.

AI evolution can therefore be understood as an advancing approximation frontier whose composition is repeatedly altered by the progress already made. The development problem is not simply the same problem becoming smaller; under the mechanisms examined here, **it can change in kind as well as scale.**

Appendix A. Compact Approximation Notation

The main text uses approximation as a functional idea rather than as a scalar measure of intelligence. The following notation records that distinction compactly.

Let $\mathbf{HI}_t$ denote the manifestations of human intelligence relevant to the domain under consideration at time t :

$$\mathbf{HI}_t = \mathbf{F}_t(\mathbf{X}_t)$$

where $\mathbf{X}_t$ denotes the human, social, environmental, institutional, embodied, and other conditions relevant to those manifestations, and $\mathbf{F}_t(\cdot)$ denotes the incompletely understood process through which those conditions generate them.

Let $\mathbf{AI}_t$ denote the corresponding artificial-system capability at time t :

$$\mathbf{AI}_t = \mathbf{G}_t(\mathbf{I}_t)$$

where $\mathbf{G}_t(\cdot)$ is the artificial mechanism then available and $\mathbf{I}_t$ is its information and computational environment, including data, feedback, tools, interaction, synthetic material, and computational resources.

$\mathbf{HI}_t$ and $\mathbf{AI}_t$ denote capability profiles or relevant manifestations, not one-dimensional intelligence scores. The notation marks an epistemic asymmetry: AI development can improve $\mathbf{G}_t$ and $\mathbf{I}_t$ without first recovering the full human-generating process $\mathbf{F}_t$.

Appendix B. A Prospective Test of Differential Resolution

The differential-resolution hypothesis can be examined without defining difficulty by the fact that a problem remained unresolved.

A prospective design could use development problems or evaluation dimensions as units of analysis. Before observing later outcomes, the study would code target specifiability, evaluability, and feedback availability using a fixed rubric. It would then define resolution using a common threshold and compare time-to-resolution over a pre-specified observation window.

The design would also record effort allocation, benchmark changes, and major shifts in available data or compute. Those variables matter because observed resolution rates reflect both tractability and the resources directed toward a problem.

The purpose of this design is not to produce a single universal ranking of AI-development difficulty. It is to test the narrower proposition advanced in Section 3: whether independently coded differences in specifiability, evaluability, and feedback availability systematically predict how quickly otherwise comparable development problems are resolved.

AI-Use Disclosure

Generative AI tools were used during the development of this manuscript for research assistance, critique, drafting support, and editorial refinement. The author independently evaluated the resulting suggestions and takes full responsibility for the manuscript’s arguments, sources, and final content.

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